

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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1. TITLE

Design and Style a Webpage Using CSS3 and the HTML5 Canvas Element

2. AIM

To design a styled web page using **CSS3**, demonstrating selectors, the box model, gradients, shadows, transitions and hover effects, flexbox layout and responsive design, and to draw graphics on an **HTML5** `<canvas>` element using JavaScript.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the three ways of applying CSS — inline, internal and external style sheets.
 - To apply CSS selectors: element selectors, class selectors, ID selectors and descendant selectors.
 - To implement the CSS box model using margin, padding, border and border-radius.
 - To apply CSS3 visual effects such as linear gradients, box-shadow, text-shadow, transitions and transform.
 - To use pseudo-class selectors such as `:hover` to create interactive effects.
 - To build a flexible card layout using the flexbox model (display: flex, flex-wrap, justify-content).
 - To control typography using font-family, font-size and color properties.
 - To draw graphics — a rectangle, a circle, a line and text — on an HTML5 `<canvas>` using the 2D rendering context.
 - To write an external style sheet using CSS variables, CSS Grid and media queries for responsive design.
-

4. THEORY

CSS (Cascading Style Sheets) is the language used to describe the presentation of an HTML document — its colours, fonts, spacing and layout. Separating presentation from structure makes a website easier to maintain and allows the same styling to be reused across many pages. CSS can be applied in three

ways: inline (using the style attribute), internal (using a `<style>` block inside `<head>`), and external (using a separate .css file linked with `<link rel="stylesheet">`).

Selectors decide which elements a rule applies to. An element selector such as `body` or `nav` targets all elements of that type, a class selector such as `.card` targets all elements carrying that class, and an ID selector such as `#myCanvas` targets a single unique element. Descendant selectors such as `.card h3` target an element only when it appears inside another.

The Box Model treats every element as a rectangular box made up of content, padding, border and margin. The `border-radius` property rounds the corners of the box and `box-shadow` draws a shadow around it, giving the card a raised appearance.

CSS3 Visual Effects — `linear-gradient()` blends two or more colours smoothly across an element, `text-shadow` adds depth to text, `transition` animates a property change over a given duration, and `transform: scale()` enlarges an element. Together with the `:hover` pseudo-class, they produce smooth interactive feedback when the mouse moves over a navigation link or a button.

Flexbox is a one-dimensional layout model. Setting `display: flex` on a container lets its children be distributed along a line; `justify-content` controls the spacing between them and `flex-wrap` allows them to move on to the next line when the window becomes narrow, which makes the card layout responsive.

The HTML5 Canvas element provides a rectangular drawing surface that is scripted with JavaScript. The element itself is only a container; all drawing is done through the rendering context obtained with `getContext("2d")`. The main drawing operations used in this experiment are:

- `fillStyle` — sets the colour used to fill shapes and text.
- `fillRect(x, y, width, height)` — draws a filled rectangle.
- `beginPath()` — starts a new path before drawing a shape.
- `arc(x, y, radius, startAngle, endAngle)` — draws a circular arc; a full circle uses 0 to $2 \times \text{Math.PI}$.
- `fill()` — fills the current path with the fill colour.
- `moveTo(x, y)` and `lineTo(x, y)` — define the end points of a line.
- `strokeStyle` and `lineWidth` — set the colour and thickness of the outline.
- `stroke()` — draws the outline of the current path.
- `font` and `fillText(text, x, y)` — draw text on the canvas.

Responsive Design is achieved with media queries. A rule such as `@media (max-width: 768px)` applies its styles only when the viewport is narrower than 768 pixels, allowing the layout to adapt to mobile screens. **CSS variables**, declared inside the `:root` selector and read with `var()`, store reusable values such as the theme colour in one place. **CSS Grid**, with `grid-template-columns` and `repeat(auto-fit, minmax(...))`, builds two-dimensional layouts that automatically adjust the number of columns to the available width.

5. ALGORITHM / PROCEDURE

- Create a new file named `My_Styled_Webpage.html`.
- Add the `<!DOCTYPE html>` declaration and the `<html lang="en">` root element.
- Inside `<head>`, add the charset meta tag, the viewport meta tag and the page title.
- Add an internal `<style>` block inside `<head>` to hold all the CSS rules.
- Style the body with a linear gradient background and a default font family.
- Style the `<header>` with a green linear gradient, white centred text, padding and a text-shadow.

- Style the `<nav>` bar with a dark background and centred links, and remove the underline with `text-decoration: none`.
- Add a `nav a:hover` rule that inverts the colours and rounds the corners to create a hover effect.
- Create a `.canvas-section` class that centres the canvas, and give the canvas a border and a white background.
- Create a `.container` class with `display: flex`, `justify-content: space-around` and `flex-wrap: wrap`.
- Create a `.card` class with a fixed width, border, border-radius, margin, padding and box-shadow.
- Style `.card h3` and `.card p` with different font families, sizes and colours.
- Style the `<button>` element and add a `button:hover` rule using `transform: scale()` and a transition.
- Style the `<footer>` with a dark background and centred white text.
- In the `<body>`, add the header, the navigation bar with four links, and the canvas section containing `<canvas id="myCanvas" width="500" height="300">`.
- Add a container `<div>` holding three card `<div>`s, each with a heading, a paragraph and a button.
- Add the footer with the copyright notice.
- At the end of the body, add a `<script>` block; obtain the canvas element with `getElementById()` and its context with `getContext("2d")`.
- Draw a filled rectangle using `fillStyle` and `fillRect()`.
- Draw a filled circle using `beginPath()`, `arc()` and `fill()`.
- Draw a line using `beginPath()`, `moveTo()`, `lineTo()`, `strokeStyle`, `lineWidth` and `stroke()`.
- Draw the student name on the canvas using `font` and `fillText()`.
- Create a separate external style sheet `style.css` demonstrating CSS variables, grid layout and media queries.
- Save the files and open the HTML file in a browser to verify the output.

6. PROGRAM

My_Styled_Webpage.html

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My Styled Webpage</title>

    <style>

        /* ===== */
        /* GENERAL STYLING                               */
        /* ===== */

        body {
            margin: 0;
            font-family: Arial, sans-serif;
            background: linear-gradient(to right, #dfe9f3, #ffffff);
        }

        /* ===== */
        /* HEADER                                           */
        /* ===== */

        header {
            background: linear-gradient(to right, #4CAF50, #2E8B57);
            color: white;
            text-align: center;
            padding: 20px;
            font-size: 35px;
            text-shadow: 2px 2px 5px black;
        }
    </style>

```

```

/* ===== */
/* NAVIGATION BAR */
/* ===== */
nav {
    background-color: #333;
    padding: 15px;
    text-align: center;
}
nav a {
    color: white;
    text-decoration: none;
    margin: 15px;
    font-size: 18px;
    padding: 8px 15px;
    transition: 0.3s;
}
/* Hover Effect for Navigation */
nav a:hover {
    background-color: white;
    color: black;
    border-radius: 5px;
}
/* ===== */
/* CANVAS SECTION */
/* ===== */
.canvas-section {
    text-align: center;
    margin: 30px;
}
canvas {
    border: 2px solid black;
    background-color: white;
}
/* ===== */
/* CONTENT CARDS */
/* ===== */
.container {
    display: flex;
    justify-content: space-around;
    margin: 30px;
    flex-wrap: wrap;
}
.card {
    width: 250px;
    background-color: white;
    border: 2px solid #ddd;
    border-radius: 12px;
    margin: 20px;
    padding: 20px;
    text-align: center;
    box-shadow: 5px 5px 12px gray;
}
.card h3 {
    font-family: Georgia, serif;
    font-size: 24px;
    color: darkblue;
}
.card p {
    font-size: 16px;
    font-family: Verdana, sans-serif;
}
/* ===== */
/* BUTTON */
/* ===== */
button {
    background-color: green;
    color: white;
    padding: 10px 20px;
    border: none;
    border-radius: 8px;
    cursor: pointer;
}

```

```

        transition: 0.3s;
    }
    /* Hover Effect for Button */
    button:hover {
        background-color: darkgreen;
        transform: scale(1.05);
    }
    /* ===== */
    /* FOOTER */
    /* ===== */
    footer {
        background-color: #333;
        color: white;
        text-align: center;
        padding: 20px;
        margin-top: 30px;
    }
</style>
</head>
<body>
    <!-- ===== -->
    <!-- HEADER -->
    <!-- ===== -->
    <header>
        My Styled Webpage
    </header>
    <!-- ===== -->
    <!-- NAVIGATION BAR -->
    <!-- ===== -->
    <nav>
        <a href="#">Home</a>
        <a href="#">About</a>
        <a href="#">Services</a>
        <a href="#">Contact</a>
    </nav>
    <!-- ===== -->
    <!-- HTML5 CANVAS -->
    <!-- ===== -->
    <section class="canvas-section">
        <h2>HTML5 Canvas Example</h2>
        <canvas id="myCanvas" width="500" height="300"></canvas>
    </section>
    <!-- ===== -->
    <!-- THREE CONTENT CARDS -->
    <!-- ===== -->
    <div class="container">
        <div class="card">
            <h3>Card 1</h3>
            <p>
                This card demonstrates borders, padding,
                margin, border radius and box shadow.
            </p>
            <button>Read More</button>
        </div>
        <div class="card">
            <h3>Card 2</h3>
            <p>
                CSS3 provides beautiful styling features
                like gradients and hover effects.
            </p>
            <button>Explore</button>
        </div>
        <div class="card">

```

```

        <h3>Card 3</h3>

        <p>
            HTML5 Canvas allows drawing graphics
            using JavaScript.
        </p>

        <button>Learn More</button>

    </div>

</div>

<!-- ===== -->
<!-- FOOTER -->
<!-- ===== -->

<footer>

    &copy; 2026 My Styled Webpage | Designed using HTML5 & CSS3

</footer>

<!-- ===== -->
<!-- JAVASCRIPT FOR CANVAS -->
<!-- ===== -->

<script>

    var canvas = document.getElementById("myCanvas");
    var ctx = canvas.getContext("2d");

    /* Rectangle */

    ctx.fillStyle = "skyblue";
    ctx.fillRect(30, 30, 120, 70);

    /* Circle */

    ctx.beginPath();
    ctx.arc(250, 65, 40, 0, 2 * Math.PI);
    ctx.fillStyle = "orange";
    ctx.fill();

    /* Line */

    ctx.beginPath();
    ctx.moveTo(50, 180);
    ctx.lineTo(350, 180);
    ctx.strokeStyle = "red";
    ctx.lineWidth = 3;
    ctx.stroke();

    /* Name */

    ctx.font = "24px Arial";
    ctx.fillStyle = "blue";
    ctx.fillText("R.sandeep", 130, 260);

</script>

</body>

</html>

```

style.css (external style sheet — CSS variables, grid layout and media queries)

```

/* Base Resets & Variables */
:root {
    --primary-color: #ff4757;
    --primary-hover: #ff6b81;
    --dark-color: #2f3542;
    --light-bg: #f1f2f6;
    --white: #ffffff;
    --text-muted: #747d8c;
    --transition: all 0.3s ease;
}

* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: 'Poppins', sans-serif;
    color: var(--dark-color);
    line-height: 1.6;
    background-color: var(--white);
}

```

```

        scroll-behavior: smooth;
    }
/* Reusable Components */
.container {
    width: 90%;
    max-width: 1200px;
    margin: 0 auto;
}

.btn {
    display: inline-block;
    padding: 10px 24px;
    border-radius: 30px;
    text-decoration: none;
    font-weight: 600;
    font-size: 0.95rem;
    transition: var(--transition);
    cursor: pointer;
    border: none;
}

.btn-primary {
    background-color: var(--primary-color);
    color: var(--white);
}

.btn-primary:hover {
    background-color: var(--primary-hover);
}

.btn-secondary {
    background-color: transparent;
    color: var(--dark-color);
}

.btn-secondary:hover {
    color: var(--primary-color);
}

.section-title {
    text-align: center;
    font-size: 2rem;
    margin-bottom: 3rem;
    position: relative;
}

.section-title::after {
    content: '';
    position: absolute;
    bottom: -10px;
    left: 50%;
    transform: translateX(-50%);
    width: 60px;
    height: 4px;
    background-color: var(--primary-color);
    border-radius: 2px;
}

/* Header & Navbar */
.header {
    background-color: var(--white);
    box-shadow: 0 2px 10px rgba(0, 0, 0, 0.05);
    position: sticky;
    top: 0;
    z-index: 1000;
    padding: 15px 0;
}

.navbar {
    display: flex;
    justify-content: space-between;
    align-items: center;
}

.logo {
    font-size: 1.8rem;
    font-weight: 700;
    color: var(--dark-color);
}

.logo span {
    color: var(--primary-color);
}

```

```

.nav-links {
  display: flex;
  align-items: center;
  gap: 20px;
}

.nav-links a:not(.btn) {
  text-decoration: none;
  color: var(--dark-color);
  font-weight: 600;
  transition: var(--transition);
}

.nav-links a:not(.btn):hover {
  color: var(--primary-color);
}

/* Hero Section */
.hero {
  background: linear-gradient(rgba(0,0,0,0.5), rgba(0,0,0,0.5)),
    url('https://images.unsplash.com/photo-1504674900247-0877df9cc836?auto=format&fit=crop
&w=1600&q=80') no-repeat center center/cover;
  height: 80vh;
  display: flex;
  align-items: center;
  color: var(--white);
}

.hero-content {
  max-width: 650px;
}

.hero h1 {
  font-size: 3.5rem;
  line-height: 1.2;
  margin-bottom: 1.5rem;
}

.hero h1 span {
  color: var(--primary-color);
}

.hero p {
  font-size: 1.2rem;
  margin-bottom: 2rem;
  opacity: 0.9;
}

/* Search Bar styling */
.search-container {
  display: flex;
  background-color: var(--white);
  padding: 8px;
  border-radius: 50px;
  box-shadow: 0 4px 15px rgba(0,0,0,0.15);
}

.search-container input {
  flex: 1;
  border: none;
  padding: 12px 20px;
  font-size: 1rem;
  border-radius: 50px;
  outline: none;
}

.search-container button {
  font-size: 1rem;
}

/* How It Works Section */
.steps-section {
  padding: 5rem 0;
  background-color: var(--light-bg);
}

.steps-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(280px, 1fr));
  gap: 30px;
}

.step-card {
  background-color: var(--white);
  padding: 40px 30px;
}

```

```

        border-radius: 12px;
        text-align: center;
        box-shadow: 0 5px 15px rgba(0,0,0,0.02);
        transition: var(--transition);
    }

    .step-card:hover {
        transform: translateY(-5px);
        box-shadow: 0 10px 20px rgba(0,0,0,0.08);
    }

    .step-icon {
        font-size: 3rem;
        margin-bottom: 20px;
    }

    .step-card h3 {
        margin-bottom: 15px;
        font-size: 1.3rem;
    }

    .step-card p {
        color: var(--text-muted);
    }

    /* Restaurants Section */
    .restaurants-section {
        padding: 5rem 0;
    }

    .restaurant-grid {
        display: grid;
        grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));
        gap: 30px;
    }

    .restaurant-card {
        background-color: var(--white);
        border-radius: 12px;
        overflow: hidden;
        box-shadow: 0 4px 15px rgba(0,0,0,0.05);
        transition: var(--transition);
    }

    .restaurant-card:hover {
        transform: translateY(-5px);
        box-shadow: 0 8px 25px rgba(0,0,0,0.1);
    }

    .card-img {
        height: 200px;
        background-size: cover;
        background-position: center;
    }

    .card-content {
        padding: 25px;
    }

    .tag {
        display: inline-block;
        padding: 4px 12px;
        border-radius: 20px;
        font-size: 0.75rem;
        font-weight: 700;
        text-transform: uppercase;
        margin-bottom: 15px;
    }

    .tag-pizza { background-color: #ffeaa7; color: #d63031; }
    .tag-burger { background-color: #fff9c4; color: #e67e22; }
    .tag-healthy { background-color: #e3fcef; color: #00b894; }

    .restaurant-card h3 {
        margin-bottom: 10px;
    }

    .delivery-info {
        font-size: 0.9rem;
        color: var(--text-muted);
        margin-bottom: 20px;
    }

    .card-link {
        text-decoration: none;
        color: var(--primary-color);
    }

```

```

        font-weight: 600;
        transition: var(--transition);
    }
    .card-link:hover {
        letter-spacing: 0.5px;
    }
    /* Footer Section */
    .footer {
        background-color: var(--dark-color);
        color: var(--white);
        padding: 30px 0;
        text-align: center;
    }
    .footer-logo {
        font-size: 1.8rem;
        font-weight: 700;
        margin-bottom: 10px;
    }
    .footer-logo span {
        color: var(--primary-color);
    }
    .footer p {
        color: var(--text-muted);
        font-size: 0.9rem;
    }
    /* Responsive Media Queries */
    @media (max-width: 768px) {
        .nav-links {
            display: none; /* Simplifies layout for mobile viewing */
        }
        .hero h1 {
            font-size: 2.5rem;
        }
        .search-container {
            flex-direction: column;
            background: transparent;
            box-shadow: none;
            gap: 10px;
            padding: 0;
        }
        .search-container input {
            background-color: var(--white);
            border-radius: 30px;
            box-shadow: 0 4px 10px rgba(0,0,0,0.1);
        }
        .search-container button {
            border-radius: 30px;
            padding: 14px;
        }
    }
}

```

7. CODE EXPLANATION

Code	Explanation
<style>	Holds the internal style sheet inside the <head> section.
body { }	Sets the page font and a linear-gradient background.
header { }	Applies a gradient background, white text, padding and a text shadow.
nav a { }	Styles the navigation links and removes their underline.
nav a:hover	Changes the link colours and rounds the corners on mouse-over.

Code	Explanation
.canvas-section	Class selector that centres the canvas on the page.
canvas { }	Gives the drawing surface a border and a white background.
.container	Uses display: flex and flex-wrap to arrange the cards in a responsive row.
.card	Demonstrates the box model — border, border-radius, margin, padding and box-shadow.
.card h3 / .card p	Descendant selectors that set the fonts, sizes and colours inside a card.
button { }	Styles the buttons with a background colour, padding and rounded corners.
button:hover	Enlarges the button using transform: scale() with a smooth transition.
footer { }	Styles the footer with a dark background and centred white text.
<canvas>	Creates a 500 x 300 pixel drawing surface with the id myCanvas.
getElementById()	Fetches the canvas element from the document.
getContext("2d")	Returns the 2D drawing context used for all canvas operations.
fillStyle / fillRect()	Set the fill colour and draw the sky-blue rectangle.
beginPath() / arc()	Start a new path and draw the orange circle.
moveTo() / lineTo()	Define the start and end points of the red line.
strokeStyle / lineWidth	Set the colour and thickness of the line.
stroke()	Draws the outline of the current path.
font / fillText()	Draw the student name in blue text on the canvas.
:root { --var }	Declares reusable CSS variables in the external style sheet.
grid-template-columns	Creates a responsive grid that fits as many columns as possible.
@media (max-width)	Applies alternate styles when the screen is narrower than 768 pixels.

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Open the HTML file in a web browser	Styled webpage should be displayed correctly	Styled webpage displayed correctly	Pass
TC-02	Observe the header and body background	Linear gradient backgrounds and text shadow should be visible	Gradients and text shadow displayed	Pass

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-03	Move the mouse over a navigation link	Link background should turn white with black text and rounded corners	Hover effect applied	Pass
TC-04	Observe the canvas area	Rectangle, circle, line and the name should be drawn	All shapes and text drawn correctly	Pass
TC-05	Observe the three content cards	Cards should show borders, rounded corners and box shadows	Box model and shadow applied	Pass
TC-06	Move the mouse over a button	Button should darken and enlarge smoothly	Transition and scale effect applied	Pass
TC-07	Reduce the width of the browser window	Cards should wrap on to the next line	Flex-wrap layout adjusted correctly	Pass
TC-08	Apply the external style sheet and resize below 768px	Media query styles should take effect	Responsive styles applied	Pass

9. OUTPUT

Output 1: Styled Header, Navigation Bar, Canvas Graphics and Content Cards

8/4/26, 12:10 PM

My Styled Webpage

My Styled Webpage

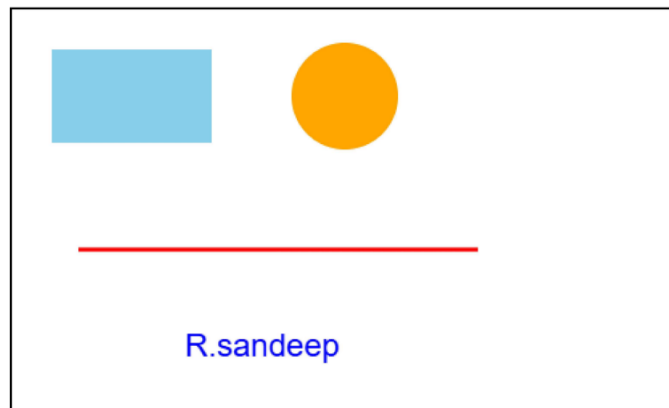
Home

About

Services

Contact

HTML5 Canvas Example



Card 1

This card demonstrates borders, padding, margin, border radius and box shadow.

[Read More](#)

Card 2

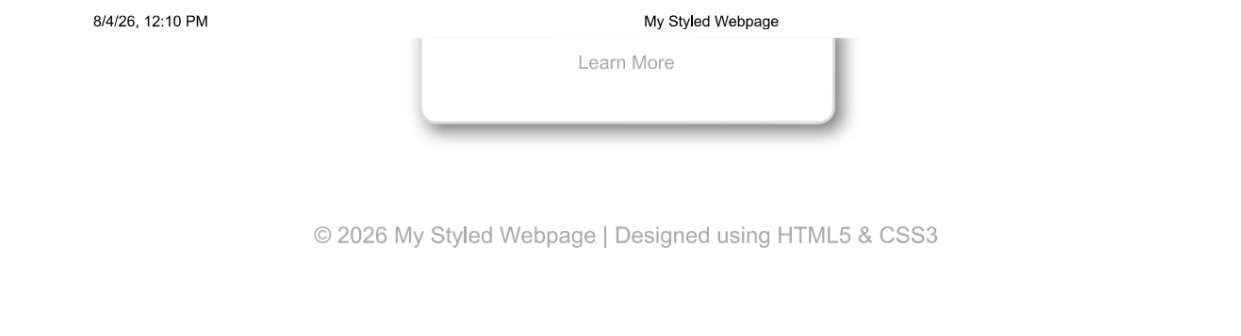
CSS3 provides beautiful styling features like gradients and hover effects.

[Explore](#)

Card 3

HTML5 Canvas allows drawing graphics using JavaScript.

Output 2: Third Content Card and Styled Footer



10. RESULT

Thus, a styled web page was successfully designed and developed using CSS3 and the HTML5 Canvas element. The webpage was verified for its gradient backgrounds, text and box shadows, box-model styling of the content cards, hover effects and transitions on the navigation links and buttons, the flexbox layout, the responsive rules of the external style sheet, and the rectangle, circle, line and text drawn on the canvas using JavaScript. All test cases were executed successfully.